DSA Mid Semester Lab Examination

Time : 75 Minutes , 15 minutes for submission

Submission Link :- https://forms.gle/nx5R5Fms3s3GwH4L9

Problem B.1: Matrix Symmetry and Identity Check Using Pointers (15 Marks)

Problem Statement: Given a square matrix of size $n \times n$, determine whether the matrix is symmetric. If it is symmetric, check whether it is also an identity matrix. Implement the solution using pointer-based traversal in C.

Input:

- The first line contains a single integer n ($1 \le n \le 500$) — the dimension of the matrix.
- The next n lines each contain n space-separated integers m_{ij} ($-10^5 \le m_{ij} \le 10^5$).

Output:

- If the matrix is not symmetric, print `NOT SYMMETRIC`.
- If the matrix is symmetric but not an identity matrix, print `SYMMETRIC`.
- If the matrix is both symmetric and an identity matrix, print `IDENTITY`.

Examples:

Example 1	Example 2	Example 3
Input: 3 1 0 0 0 1 0 0 0 1 Output: IDENTITY	Input: 3 2 1 3 1 2 1 3 1 2 Output: SYMMETRIC	Input: 3 1 2 3 4 5 6 7 8 9 Output: NOT SYMMETRIC

Problem B.2: Loop Detection in a Linked List (15 Marks)

Problem Statement: Given a singly linked list, determine whether there exists a cycle (loop) in the list or not.

Input:

- The first line contains an integer n ($1 \leq n \leq 10^5$) — the number of nodes in the linked list.
- The second line contains n space-separated integers representing the node values.
- The third line contains a single integer pos ($-1 \leq pos < n$).
  - If $pos = -1$, it means there is no cycle.
  - If $pos \geq 0$, it indicates that the next pointer of the last node points to the node at index pos (0-based), forming a cycle.

Output:

- Print `LOOP` if a cycle exists in the linked list; otherwise, print `NO LOOP`.

Example 1:

```
Input:
5
1 2 3 4 5
-1

Output:
NO LOOP
```

Example 2:

```
Input:
5
1 2 3 4 5
2

Output:
LOOP
```